ModelIt!

Student Activity Pack

Trophic Cascades: When Predators Disappear

How Removing One Species Can Collapse an Entire Ecosystem

Grade 08L2 | NGSS MS-LS2-1, MS-LS2-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is an apex predator and can you name some examples?

2 What do you think would happen if all predators in an area disappeared?

3 How are different species in an ecosystem connected to each other?

4 Draw a simple food web and predict what happens when you remove the top predator.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Apex Predator Population | Herbivore Population | Vegetation Health | Erosion Rate | Ecosystem Stability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced Ecosystem

Natural predator-prey balance

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Predator Removal

Set Apex Predator Population to 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Predator Reintroduction

Restore predators to 50% after removal

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Scavenger Population — Animals like ravens and beetles that feed on predator kills

Stream Health — Water quality and bank stability affected by vegetation

Bird Diversity — Number of bird species, which depends on habitat variety from vegetation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens to an entire ecosystem when the top predator vanishes? Use evidence from your simulation to support your answers.

1. **Why is it surprising that wolves can affect the course of rivers?**

2. **How does the Yellowstone wolf story challenge the idea that predators are just 'dangerous animals'?**

3. **Why did the ecosystem NOT recover on its own after 70 years without wolves?**

4. **What does the trophic cascade model reveal about the idea that 'everything is connected' in nature?**

5. **Should humans actively manage wildlife populations, or should we let nature find its own balance?**

8 STEM Challenge: Design a Wildlife Management Plan

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YOUR ENGINEERING MISSION

Design an evidence-based wildlife management plan for a national park experiencing ecosystem decline due to predator loss.

THE SCENARIO

A state park has removed wolves 20 years ago due to rancher complaints. Now the park is experiencing severe erosion, declining bird populations, and degraded streams. Your team must propose a recovery plan using model evidence.

Engineering Design Thinking

1. HOW MANY PREDATORS SHOULD BE REINTRODUCED AND ON WHAT SCHEDULE?
2. WHAT MONITORING SYSTEMS WILL TRACK HERBIVORE POPULATIONS AND VEGETATION RECOVERY?
3. HOW WILL YOU MANAGE CONFLICTS BETWEEN WILDLIFE AND LOCAL COMMUNITIES?
4. WHAT MILESTONES WILL INDICATE YOUR PLAN IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Ecologists and Conservation Managers study predator-prey dynamics and design management strategies for national parks and wildlife reserves. They earn $55,000-$95,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In your trophic cascade model, removing the Apex Predator Population caused Erosion Rate to spike. Which chain of cause-and-effect relationships from the model explains this?

- **A.** No predators caused soil to become drier, leading to erosion
- **B.** No predators led to herbivore explosion, which overgrazed vegetation, which exposed bare soil to erosion
- **C.** No predators caused direct physical damage to soil through increased animal movement
- **D.** No predators meant fewer decomposers, which weakened the soil structure

2 Your model shows that after predator reintroduction at 50%, ecosystem recovery took significantly longer than the original collapse. What does this asymmetry demonstrate?

- **A.** Predators are ineffective at controlling herbivore populations once they have grown too large
- **B.** Degraded ecosystems can become trapped in unhealthy stable states where damage is self-reinforcing, making recovery slower than collapse
- **C.** The model is inaccurate because recovery should take the same amount of time as collapse
- **D.** Herbivores evolved resistance to predation during the absence of predators

3 A student argues that removing wolves from a park should only affect elk, since wolves only eat elk. Using your model, which evidence contradicts this claim?

- **A.** Wolves occasionally eat plants, which directly affects vegetation
- **B.** Removing predators triggers a cascade through herbivore, vegetation, erosion, and ecosystem stability components, affecting the entire system
- **C.** Wolves are decomposers that recycle nutrients in the soil
- **D.** Wolves control the weather patterns that affect vegetation growth

4 In the model, Vegetation Health has a POSITIVE relationship with Ecosystem Stability and a NEGATIVE relationship with Erosion Rate. If Vegetation Health drops to 20%, what does the model predict?

- **A.** Ecosystem Stability increases because there is less competition among plants
- **B.** Erosion Rate decreases because there are fewer roots disrupting the soil
- **C.** Erosion Rate increases dramatically while Ecosystem Stability declines, creating compounding damage
- **D.** Both Erosion Rate and Ecosystem Stability remain unchanged because they are independent of vegetation

5 **Based on your model data, a wildlife manager must choose between two strategies for a park with no predators. Which strategy does the model evidence best support?**

- **A.** Plant new vegetation to directly restore soil cover and stop erosion
- **B.** Reintroduce apex predators to restore top-down control, which will cascade through herbivores, vegetation, and erosion over time
- **C.** Reduce erosion with artificial barriers, which will allow vegetation to regrow naturally
- **D.** Remove herbivores entirely so vegetation can recover without grazing pressure